Instructions

1. Use this slide template as the pitch deck you will submit, please **do NOT** add any additional slides. Any additional slides may not be reviewed as part of your submission. You should have **6 slides total.**
2. The “Our Team” slide, slide 3, needs to be completely filled out.
3. Reminder- you must also submit:
    a. Your squad environment link
    b. Your demo / live app link
    c. Your Github repo link
    d. Recorded pitch video, under 3 minutes. **Please record yourselves briefly explaining/pitching your solution.** You can record this from any device.



Google Cloud

Agents for Impact

Hackathon



Cooling Hotlanta

Our Team

Filming Availability: Should your team be selected as the winner, indicate if and when you are able to return to the Coca-Cola Roxy between 1:00-4:00 PM on Thursday 10/29 to film a team pitch of your project.
Project Sharing: Mark “yes” if you are OK with us sharing what you built with your team at work.
Share With: If “yes” to the above, is there anyone specific you’d like us to share it with at work? Provide email(s).
👉 **Check your Discord channel tonight to learn if you won. We will contact you no later than 9:00 PM.**

NAME	EMAIL	PHONE	FILMING AVAILABILITY (between 1:00-4:00 PM on 10/29)	PROJECT SHARING	SHARE WITH (provide emails)
Tanya Huggler	tanya.huggler@coxautoinc.com	518-542-5335	1:00		
Agustin Corigliano	pagustincorigliano@gmail.com	786-721-3109	1:00		
Joel Metuge	joelmetuge0707@gmail.com	256-797-2970	1:00		
David Ballagh	david.ballagh@gmail.com	352-978-8537	1:00		
Vamshi Gangula	vkgangula@gmail.com	503-726-8746	1:00		

Challenge Overview

Real-world challenge
Severe urban heat islands threatening public health across vulnerable, tree-deprived Atlanta neighborhoods.

Details
Vulnerable residents benefit; mitigates heat deaths, lowers energy costs, and advances climate equity.

Impact

Our Agent
Predicts hyper-local heat spikes to deploy targeted cooling interventions and life-saving community alerts.

Details
Planners query interactive heat maps to prioritize shade infrastructure, optimizing municipal budgets and resilience.

Sources



Our data ecosystem is orchestrated on Google Cloud, fusing petabyte-scale geospatial telemetry, municipal infrastructure layers, and public health indicators inside BigQuery.

By coupling Google Earth Engine's automated satellite pipeline with BigQuery Spatial SQL, the agent performs real-time multi-criteria analysis across environmental, structural, and demographic variables.

This continuous data backbone breaks down municipal silos, transforming raw static archives into a live predictive intelligence engine tailored for municipal decision-makers.

Details

We ingest high-resolution Landsat-9 and Sentinel-2 thermal imagery alongside live NOAA weather stations to map surface temperature anomalies down to the parcel level.

These thermal models are joined with City of Atlanta Open GIS layers—specifically LiDAR tree canopy assessments, impervious surface footprints, and municipal zoning boundaries—to diagnose built-environment heat traps.

Crucially, the agent cross-references CDC Social Vulnerability Index (SVI) BigQuery tables, U.S. Census ACS economic data, and NREL energy burden metrics to ensure cooling investments are prioritized in historically underserved neighborhoods.

*Architecture included in GitHub

Prototype Architecture

Details: Multi-Agent Mechanics & Communication

1. **Supervisor Orchestration (ADK Framework):**
   The **Lead Orchestrator Agent** acts as the central state manager built using the **Agent Development Kit (ADK)**. When a planner inputs a goal (e.g., *"Allocate $2M to reduce heat vulnerability in District 4"*), the Orchestrator decomposes the request into parallel, task-specific sub-queries and manages context across sub-agents.

2. **Standardized Tool Integration (Model Context Protocol / MCP):**
   Each sub-agent connects to underlying data lakes using standardized **MCP servers via JSON-RPC**:

   - **Thermal Agent:** Calls the Earth Engine MCP tool to retrieve real-time Landsat-9 Land Surface Temperatures (LST) and NOAA meteorological data.
   - **Urban Spatial Agent:** Calls the BigQuery Spatial MCP tool to evaluate Atlanta parcel footprints, LiDAR tree canopy, and impervious asphalt ratios.
   - **Social Vulnerability Agent:** Interfaces with BigQuery to extract CDC SVI metrics and Census demographic profiles.

3. **Synthesis & Feedback Loop:**
   Once data agents return their vector analyses, the **Scenario Optimizer Agent** ingests the combined context to simulate high-impact interventions (e.g., cool roofs, reflective pavement, shade trees) constrained by municipal budgets. The Orchestrator validates the solution against city policy constraints and exports an actionable GeoJSON map layer and prioritized investment report directly back to the planner.